Postural Influence on Swahili Vowel Characteristics

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April 29, 2026

1 Rationale

This project consists of a hands-on small-scale speech corpus construction for the Swahili language. The objective is threefold: the collection of speech data through recording thirty audio files across three positions (standing, sitting, and supine), a two-layer manual annotation focusing on words and phonemes using Praat software, and the statistical investigation of differences in vowel characteristics for the phonemes /a, e, i, o, u/.

2 Data Collection

Audio samples were recorded on an Android phone using the mobile application *Audio Recorder*[1]. The recordings were captured at a sampling frequency of 22 kHz in mono and saved as WAV files without compression. Each file has a duration between 3.60s and 5.92s, with a mean duration of 4.79s. The data collection occurred across three distinct positions: standing, sitting, and supine. Ten sentences using everyday Swahili language were recorded for each position.

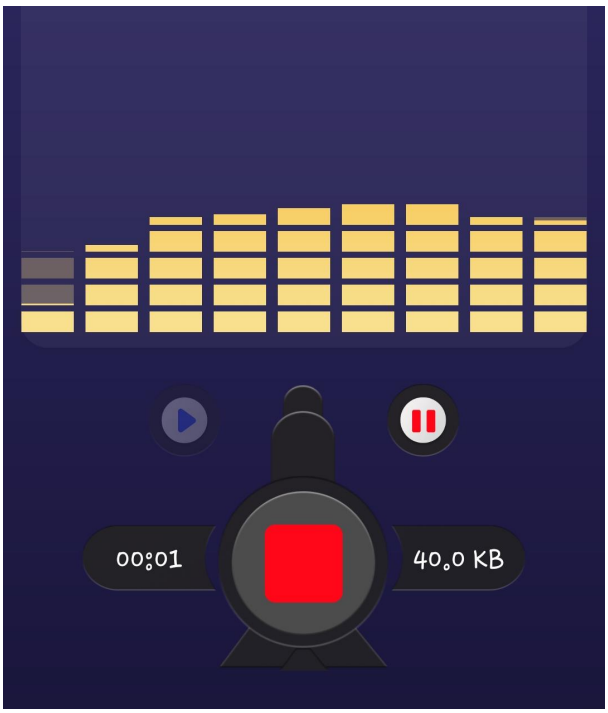

Figure 1: Interface of the Audio Recorder application used for data collection.

3 Speech Data Annotation

For annotation, a two-layer fully manual approach was adopted, focusing on phonemes and words using **Praat**[2]. Due to Swahili being a low-resource language, the limitations of existing tools like the Montreal Forced Aligner[3] were significant, necessitating manual

[1] https://play.google.com/store/apps/details?id=com.media.bestrecorder.audiorecorder
[2] https://www.fon.hum.uva.nl/praat/
[3] https://mfa-models.readthedocs.io/en/latest/dictionary/Swahili/Swahili%20MFA%20dictionary%20v2_0_0a.html

alignment for precision. Phoneme annotation followed *SAMPA notation*[4].

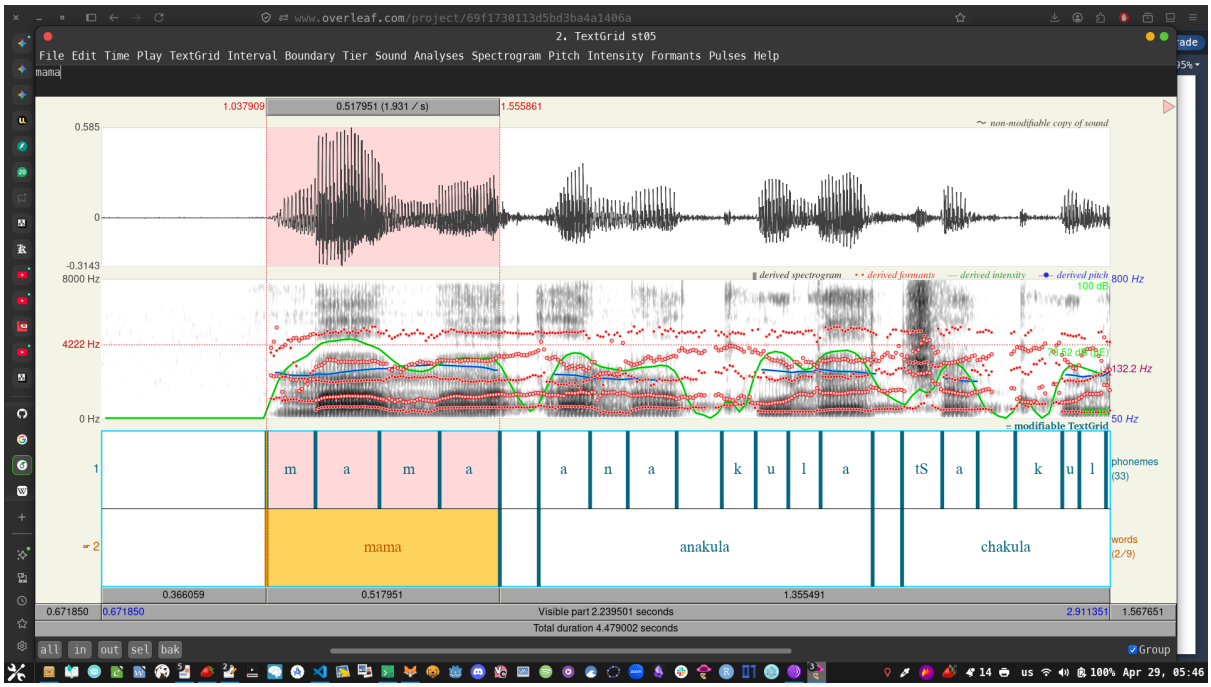

Figure 2: Manual annotation of speech data within the Praat environment.

4 Formant Extraction and Analysis

Following the pairing of audio samples with their respective TextGrid annotations, **Parselmouth**[5] was employed for formant extraction. Formants were calculated at the midpoint of each vowel to ensure acoustic stability. The source code and dataset are hosted on GitHub[6].

4.1 Formant Extraction and Preprocessing

Formant values (F_1, F_2, F_3) were computed for every vowel identified in the annotations. A preprocessing step was implemented to filter out values potentially out of range or resulting from tracking artifacts.

4.2 Summary Statistics

Mean and standard deviation values for F_1, F_2, and F_3 were calculated for the five target vowels across each position. The following tables present these descriptive statistics along with the token counts before and after the filtering process.

[4] `https://en.wikipedia.org/wiki/SAMPA_chart`
[5] `https://parselmouth.readthedocs.io/en/stable/`
[6] `https://github.com/SalomonMetre/swahili_corpus_construction`

Table 1: Formant Statistics for **Standing**

Vowel	Initial Count	Final Count	F_1 Mean (SD)	F_2 Mean (SD)	F_3 Mean (SD)
a	61	57	580.09 (65.41)	1387.27 (134.32)	2288.41 (142.78)
e	11	10	431.37 (33.94)	1938.89 (52.86)	2528.16 (59.36)
i	22	17	379.25 (73.38)	2075.43 (92.75)	2768.38 (188.97)
o	9	8	412.94 (36.88)	982.31 (56.84)	2216.71 (67.80)
u	15	13	386.41 (46.84)	1000.94 (170.99)	2210.12 (78.68)

Table 2: Formant Statistics for **Sitting**

Vowel	Initial Count	Final Count	F_1 Mean (SD)	F_2 Mean (SD)	F_3 Mean (SD)
a	61	59	573.60 (48.61)	1452.37 (135.99)	2284.27 (144.85)
e	11	10	421.32 (16.23)	1906.81 (77.25)	2443.70 (92.94)
i	22	17	345.80 (33.61)	2077.40 (157.44)	2691.32 (183.80)
o	9	8	437.49 (46.46)	995.07 (233.21)	2308.76 (86.13)
u	15	13	406.02 (54.69)	1087.17 (328.76)	2334.43 (276.42)

Table 3: Formant Statistics for **Supine**

Vowel	Initial Count	Final Count	F_1 Mean (SD)	F_2 Mean (SD)	F_3 Mean (SD)
a	61	56	592.86 (60.53)	1418.59 (170.44)	2247.60 (223.72)
e	11	10	431.08 (19.77)	1929.94 (109.25)	2585.77 (201.53)
i	22	18	366.63 (53.88)	2121.39 (91.02)	2863.45 (170.11)
o	9	8	419.53 (27.53)	974.96 (82.55)	2289.72 (89.24)
u	15	12	384.80 (37.57)	1062.87 (232.84)	2314.59 (192.42)

5 Statistical Significance of Postural Orientation on Formant Values

Statistical assessments were conducted to determine the extent to which physical orientation influences articulatory output. The analysis is divided into phoneme-specific paired tests and a global assessment of the acoustic space.

5.1 Vowel-level Analysis

Paired t-tests were executed to evaluate shifts at the phoneme level across the three postures. The analysis of the first formant (F1), which correlates with vowel height and jaw aperture, indicates that significance is exclusively observed for the vowel **A** when transitioning between sitting and supine positions (p=0.0247). No other vowels demonstrated significant F1 fluctuations across the tested orientations.

Regarding the second formant (F2), associated with vowel backness and tongue retraction, a highly significant shift is detected for the vowel **A** in the standing vs. sitting

comparison (p=0.0002). The remaining vowels (E, I, O, U) maintained stable F2 values across all postural comparisons.

The third formant (F3), which often reflects vocal tract length and lip rounding, showed significant results for several phonemes: vowel **E** in the standing vs. sitting comparison (p=0.044), vowel **I** in the sitting vs. supine comparison (p=0.0388), and vowel **O** in the standing vs. sitting comparison (p=0.0409). Apart from these specific instances, no other vowel-level comparisons reached the threshold for statistical significance.

5.2 Global Analysis

A global analysis was conducted by aggregating the data for all vowels to determine if posture exerts a systematic effect on the overall acoustic space. For all three formants (F1, F2, and F3), pairwise comparisons across the three positions resulted in p-values consistently greater than 0.05. These results indicate that, at a macro level, postural orientation does not have a significant impact on combined formant values. This suggest that the speech motor system successfully maintains global acoustic constancy despite the minor articulatory variations observed at the individual phoneme level.

6 Corpus Construction Cost

The cost of corpus construction is primarily evaluated through the labor-hours required for manual phoneme alignment. For this project, the annotation effort was divided into two phases based on software proficiency. The first five audio files required 15 minutes each, while the subsequent twenty-five files required approximately 7.5 minutes each, totaling 262.5 minutes (4.375 hours) of manual labor.

Based on the current gross hourly minimum wage in France (SMIC) of approximately 11.65€, the estimated labor cost for this small-scale corpus is 50.97€. When evaluated against the 724 annotated phonemes, the labor cost per phoneme is approximately 0.07€. This highlights the significant economic and human expertise required to develop high-quality manual datasets for low-resource languages where automated tools are not yet viable.

7 Uses of the Corpus

The resulting speech data and corresponding phonemic annotations serve as a valuable resource for several Natural Language Processing tasks. Such data is foundational for the development and fine-tuning of **Automatic Speech Recognition** (ASR) systems, particularly in improving phoneme-to-text mapping in low-resource contexts. Furthermore, the corpus provides necessary training material for **Text-to-Speech** (TTS) synthesis, where precise temporal alignment of phonemes is critical for generating natural-sounding speech.

8 References

- `https://mfa-models.readthedocs.io/en/latest/dictionary/Swahili/Swahili%20MFA%20dictionary%20v2_0_0a.html`

- McAuliffe, M., Socolof, M., Mihuc, S., Wagner, M., and Sonderegger, M. (2017). Montreal Forced Aligner: Trainable Text-Speech Alignment Using Kaldi. In *Interspeech 2017*, 498–502. doi: 10.21437/Interspeech.2017-1386.
- *Audio Recorder* Android App: `https://play.google.com/store/apps/details?id=com.media.bestrecorder.audiorecorder`
- *Praat: doing phonetics by computer*: `https://www.fon.hum.uva.nl/praat/`